Paul Holland's Insights into the Fundamental Problem of Causal Inference

Motivation, Models, and the Fundamental Problem

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Key Figures



Paul W. Holland

1940–
University of Michigan
Stanford University
Educational Testing Service (ETS)



Donald B. Rubin

1943–
Princeton University
University of Chicago
Harvard University

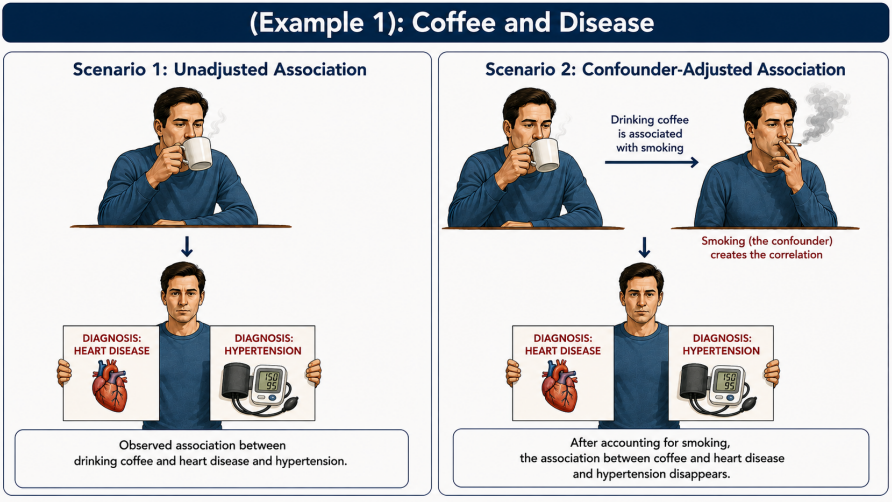
Correlation $\neq$ Causation

A statistical association alone does not imply a causal relationship.

But...

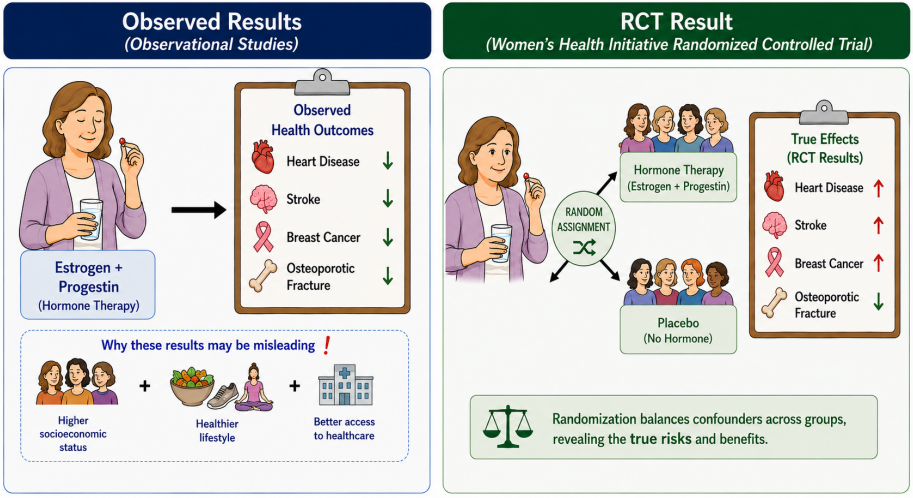
Causal conclusions from a **carefully designed randomized experiment** are often valid.

Example 1



(Schreiber et al., 1988)

Example 2



(Rossouw et al., 2002)

Associational Model

The study population is

$$U = \{u_1, u_2, \ldots, u_N\},$$

where $u \in U$ denotes a primary unit of interest.

A variable is a real-valued function

$$Y : U \to \mathbb{R},$$

so that for each unit $u \in U$, Y assigns a real-valued observation $Y(u)$.

The response variable Y is the variable of interest, and we seek to understand

How does $Y(u)$ vary across different $u \in U$?

Associational Inference

- **Associational inference**

  Statistical inference about the **associational parameters** relating Y and A.

- Based on data about Y **and** A collected from units in U.

- Includes: **Estimation, hypothesis testing, posterior distributions, etc.**

- **In this sense, associational inference is simply descriptive statistics.**

Holland's Focus on Causation

Traditional Focus

Causes of Effects

Holland's Focus

Effects of Causes

Causal model (Rubin's Model)

- Holland uses the **language and framework of experiments** to clarify causal inference.
- The model is a simplified, **population-level version of Rubin's model**.
- Applied to: **retrospective case-control studies** and **Lord's paradox**.

Rubin's Model for Causal Inference

Holland (1986); Holland & Rubin (1980, 1983)

Cause and Treatment

- **Cause := Treatment**
- We use **cause** and **treatment** interchangeably because the **effect of a cause is always relative to another cause**.
- **"A causes B"** means A causes B **relative to another cause**, typically **not A**.
- Experimental language makes this comparison clear:

Treatment (A) vs. **Control (not A)**

Causal Inference: The Counterfactual

TREATMENT (A)	NOT TREATMENT (NOT A)
Assassination of Franz Ferdinand occurs	Assassination of Franz Ferdinand does not occur
	
World War I breaks out	Would World War I have happened? (Possibly later, or not at all)

TREATMENT (A)	NOT TREATMENT (NOT A)
High exposure to Italian brainrot content	Low or no exposure to Italian brainrot content
	
Mental health outcomes (attention, mood, self-esteem, development) observed	What would their mental health outcomes have been without exposure?

What Can Be a Cause?

- **It is critical that each unit be potentially exposable to any one of the causes.**

- **The alternative causes need not be practically feasible but they must be conceptually possible.**
  - Italian brainrot exposure: A child could potentially experience high or low/no exposure.
  - Franz Ferdinand's assassination: We can consider the same historical situation with or without the assassination, even though we cannot actually go back in time (yet!!).

Causal Model (Mathematical Notation)

- **Treatment / Cause** has two levels:

$$t = \text{treatment}, \qquad c = \text{control}$$

- Population:

$$U = \{u_1, u_2, \ldots, u_N\}$$

- Define $S(u)$ to indicate the **cause** to which unit u is exposed:

$$S(u) = \begin{cases} t, & \text{unit } u \text{ receives the treatment cause,} \\ c, & \text{unit } u \text{ receives the control cause.} \end{cases}$$

Controlled study

$S(u)$ is determined by the **experimenter**.

Uncontrolled study

$S(u)$ is determined by **factors outside the experimenter's control**.

Causal Model (Mathematical Notation)

- Variables $\rightarrow$ **pre-exposure** or **post-exposure**

$$\underbrace{\text{Pre-exposure}}_{\text{value determined before exposure}} \longrightarrow \underbrace{\text{Post-exposure}}_{\text{value determined after exposure}}$$

- Let Y be the **response variable** measuring the effect of the cause.
- Since Y may depend on the cause (t, c), represent the two potential responses:

$$Y_t(u) \quad \text{and} \quad Y_c(u)$$

$$\boxed{\begin{aligned} &Y_t(u) : \text{response of } u \text{ if exposed to } t, \\ &Y_c(u) : \text{response of } u \text{ if exposed to } c. \end{aligned}}$$

- **Effect of cause t relative to c on unit u:**

$$\boxed{\tau(u) := Y_t(u) - Y_c(u)}$$

The two potential responses represent the two possible exposures.

The Fundamental Problem of Causal Inference

The Problem

For a given unit u, we can observe only **one** of the two potential responses:

$$Y_t(u) \qquad \text{or} \qquad Y_c(u)$$

It is impossible to observe both $Y_t(u)$ and $Y_c(u)$ on the same unit.

Scenario 1: $S(u) = t$

$$\underbrace{Y_t(u)}_{\text{observed}} \qquad \underbrace{Y_c(u)}_{\text{unobserved}}$$

Scenario 2: $S(u) = c$

$$\underbrace{Y_t(u)}_{\text{unobserved}} \qquad \underbrace{Y_c(u)}_{\text{observed}}$$

The causal effect on unit u cannot be observed.

Solutions to the Fundamental Problem of Causal Inference

Scientific Solution

Exploit **homogeneity / invariance**.

$Y_c(u)$ before exposure $= Y_c(u)$ at experiment

Expose u to t and observe $Y_t(u)$.

Infer the missing $Y_c(u)$.

Statistical Solution

$$U = \{u_1, u_2, \ldots, u_N\}$$

$$T = E_U[Y_t(u) - Y_c(u)]$$

$$\boxed{T = E_U[Y_t(u)] - E_U[Y_c(u)]}$$

Average causal effect over U.

The usefulness of both statistical and scientific solutions depends on the truth of a set of

untestable assumptions.

Causal Model and Observation

Causal Model

S	Y_t	Y_c

Three variables describe the two potential responses under t and c.

Observation

S	$Y_s(u)$

Only the response corresponding to the observed cause is available.

This distinction is fundamental to causal inference, but is not made in a study of simple association.

Specific assumptions added to the causal model allow causal inferences of particular types.

Methods of overcoming the fundamental problem of causal inference

Temporal Stability and Causal Transience (Scientific Solution)

Temporal Stability

$Y_c(u)$

does not depend on **when** the sequence

apply c to $u \longrightarrow$ measure Y

Causal Transience

$Y_t(u)$

is not affected by prior exposure of u to sequence

apply c to $u \longrightarrow$ measure Y

$$c \longrightarrow Y \quad \text{then} \quad t \longrightarrow Y$$

If both assumptions are valid obtain $Y_c(u)$ and $Y_t(u)$ through sequential exposure of u to c then t, measuring Y after each exposure.

Unit Homogeneity (Scientific Solution)

$$\boxed{Y_t(u_1) = Y_t(u_2), \qquad Y_c(u_1) = Y_c(u_2)}$$

for units u_1 and u_2 (**homogeneity assumption**)

$$\boxed{\text{Causal effect} = Y_t(u_1) - Y_c(u_2)}$$

Assumption Plausibility: units are prepared to be identical in all relevant aspects.

Independence (Statistical Solution)

Observed data: $\boxed{(S, Y_s)}$

$$\boxed{T := E(Y_t) - E(Y_c)}$$

The observed data give information about

$$E[Y_s \mid S = t] = E[Y_t \mid S = t], \qquad E[Y_s \mid S = c] = E[Y_c \mid S = c].$$

Independence assumption

$$\boxed{S \perp\!\!\!\perp (Y_t, Y_c)}$$

If randomization is carried out correctly, the cause assigned to u is independent of its potential responses Y_t and Y_c.

Independence (Statistical Solution)

Under independence,

$$E(Y_t) = E(Y_t \mid S = t), \qquad E(Y_c) = E(Y_c \mid S = c).$$

Therefore,

$$\boxed{T = E[Y_s \mid S = t] - E[Y_s \mid S = c]}$$

Without independence:

$$\boxed{T_{PF} = E[Y_t \mid S = t] - E[Y_c \mid S = c]}$$

T_{PF} is an **associational parameter** of the observed pair (S, Y_s).

$$\boxed{S \perp\!\!\!\perp (Y_t, Y_c) \quad \Longrightarrow \quad T = T_{PF}}$$

Constant effect assumption

The effect of t is the same for every unit:

$$\boxed{T = Y_t(u) - Y_c(u), \qquad \forall u \in U}$$

Thus, T is the causal effect for **every unit** in U.

Additivity

$$Y_t(u) = Y_c(u) + T$$

t adds the same constant amount T to the control response for every unit.

$$\boxed{\text{Constant effect}} \implies \text{unit-level causal inference}$$

Can be partially checked by dividing U into subpopulations and estimating T separately. Large differences make the assumption implausible.

Beyond the scope of this lecture

A Final Causal Question...

What is the causal effect of changing

ISRT $\longrightarrow$ IASDS

on the recognition of the institute?

What would the recognition have been under ISRT?

What assumptions would allow us to answer this?

Thank You!

Now, I would like to invite Kamruzzaman to continue the talk.

References I

Rossouw, J. E., Anderson, G. L., Prentice, R. L., LaCroix, A. Z., Kooperberg, C., Stefanick, M. L., Jackson, R. D., Beresford, S. A. A., Howard, B. V., Johnson, K. C., Kotchen, J. M., and Ockene, J. (2002). Risks and benefits of estrogen plus progestin in healthy postmenopausal women: Principal results from the women's health initiative randomized controlled trial. *JAMA*, 288(3):321–333.

Schreiber, G. B., Robins, M., Maffeo, C. E., Masters, M. N., Bond, A. P., and Morganstein, D. (1988). Confounders contributing to the reported associations of coffee or caffeine with disease. *Preventive Medicine*, 17(3):295–309.